

Label-Agnostic Representation Learning for Cross-Species Single-Cell Data

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Abstract

Narrative something like:

- Cell diversity is important (explain why, some brief examples)
- Biological datasets contain unreliable structure
- Existing models (e.g. SATURN [RBR⁺24]) use that structure as supervision
- This is potentially self-defeating, maybe limiting representation quality and scalability
- We tested unsupervised contrastive learning, as well as some distribution matching algorithms
- Label-agnostic InfoNCE [vdOLV18] produced comparable results on mammalian data
- Question on how to evaluate well still unresolved

1 Introduction

1.1 Background and motivation

- Single cell RNA sequencing
- Cell atlases
- Why learning a shared embedding space is good/useful
- Existing approaches (e.g SATURN)
- The dependence of these approaches on curated labels
- Labels may be inconsistent between datasets and species (need to find good papers showing this)
- Biological states often exist on a continuum (and this)
- Therefore, cells near an annotation boundary may be biologically similar, but treated as different by these models
- Also noise, batch effects, inconsistent annotations etc
- Recent microsoft paper [DHT⁺26]: biological models appear to not follow predictable scaling laws
- Potentially due to data quality and supervision quality

1.2 Research rationale

- If embeddings are intended to discover biological structure, it may be counterproductive to impose an uncertain and noisy structure on the embeddings before learning takes place
- Models may reproduce the existing annotation system rather than discover biological organisation
- Labels may still have value as an evaluation tool

1.3 Research problem

- Mainly: can biologically meaningful cross-species cell embeddings be learned without using cell-type labels to define the embedding space?
- Subquestions:
 - Can a label-agnostic contrastive learning objective perform comparably to supervised?
 - Can distribution-matching methods align cells across species without explicit cell-level supervision?
 - How should label-agnostic embeddings be evaluated?
 - Can broad cell categories provide a more reliable evaluation target than fine-grained cell types

2 Literature review

2.1 Cross-species single cell representation learning

- Explain single-cell gene-expression data
- Motivate the need to compare cells across species a bit
- Gene homology and evolutionary distance
- Review models (n.b. think harder about what order to put these in):
 - SAMap [TMK⁺21]
 - UCE [RRA⁺26]
 - SATURN [RBR⁺24] & SATURN-xfer [XGK⁺25]
 - TranscriptFormer [PSM⁺26]
 - Claude Mythos (claimed) [Ant26]

2.2 Supervised metric learning and its limitations

- Cluster boundary problem
- Detail why I think the SATURN algorithm is bad:
 - Semihard margin kind of fake
 - Margin is a random number
 - Positives selected by clusters
 - Look closer at old code to remember this

2.3 Label-agnostic learning objectives

2.3.1 Contrastive learning

- InfoNCE [vdOLV18]
- Positives and negatives
- Why it might work

2.3.2 Distribution matching algorithms

- MMD and MMD-MA [LHS⁺19]
- OT [Kan58]
- Why they might work
- The fact they're non-generative

2.4 Evaluating biological embeddings

- Species mixing
- Neighbourhood preservation
- Visual inspection of embeddings
- Then explain why these aren't good enough
- Introduce Fabio's broad cell categories (will need help motivating this)

3 Methodology

3.1 Research design

- Model objective
- Evaluation framework
- Diagram of the full pipeline showing SATURN into new metric learning heads into some eval

3.2 Data and preprocessing

- Data and species used (will need Fabio help with citations here)
- EDA (what is normal?)

3.3 Baseline model

- Details of SATURN
- and hyperparams and training setup

3.4 Label-agnostic models

3.4.1 InfoNCE

- How positive pairs are selected
- Negatives (i.e. everything)
- Other details

3.4.2 MMD-MA

Implementation details

3.4.3 Optimal transport

Implementation details

3.5 Evaluation

3.5.1 Biological structure

Clean-ness of categories (i.e. same neighbour fraction), visual inspection

3.5.2 Species integration

Species mixing, local neighbour composition, cross-species nearest neighbours or something

3.5.3 Model performance

Training behaviour, convergence, cost

3.6 Experiment setup

Random seeds, hardware, hyperparam, how I picked results etc, just all the other implementation details

4 Results

4.1 Overview

Method	Label-free	Category preservation	Species mixing
Supervised baseline (SATURN)	No	?	?
InfoNCE	Yes	?	?
MMD-MA	Yes	?	?
Optimal Transport	Yes	?	?

4.2 InfoNCE

- Performance vs baseline
- Details
- Visualisation

4.3 Distribution-matching results

Same as above

4.4 Comparison of learning objectives

- InfoNCE supplies a cell-level learning signal
- MMD-MA and OT align distributions instead
- Global alignment may say nothing about local alignments
- Differences in cell-type composition between species may make full distribution-matching inappropriate

5 Discussion

5.1 Summary of findings

- InfoNCE produced a representation comparable to the benchmark
- InfoNCE produced metrics and UMAP comparable to the benchmark
- Distribution matching approaches were less effective at preserving biologically meaningful structure
- Evaluating embeddings without reliance on labels remains unresolved

5.2 Interpretation and evaluation challenges

- InfoNCE did well because of xyz
- Distribution matching approaches didn't because xyz
- No reliable framework for evaluating embeddings without labels, so everything we did relied on labels one way or another
- Broad cell categories provide a partial evaluation solution via a more stable and interpretable target
- But that doesn't fully solve the problem and probably won't scale

5.3 Implications for model scaling in biology

- Limitations in biological model scaling may arise from learning signal quality, not just model capacity
- Increasing data volume without addressing annotation inconsistencies may reinforce incompatible taxonomies
- This might hinder generalisation
- We didn't directly evaluate scaling behaviour, but this highlights a possible mechanism constraining scaling

5.4 Limitations

- Highly restricted set of species (3 mammals)
- Hyperparameter exploration was limited
- All results are highly dependent on the pretrained encoder, which itself is dependent on ESM embeddings. Also the encoder doesn't produce any species mixing in itself, so this probably needs to be better
- Hard to evaluate - totally possible the final result is "wrong" in a unique way. There isn't really a rigorous external validation framework (but [CPB+18] or equivalent may help)

6 Conclusions and Future Work

6.1 Conclusions

- We investigated whether cross-species cell representations could be learned without using cell-type labels to impose the geometry of the embedding space
- InfoNCE did pretty well
- Distribution matching approaches didn't
- Evaluating these things is an open problem

6.2 Future work

- Model development
- Scaling this approach
 - Bigger/better pretraining
- Evaluation

- Build out a framework? maybe including held out biological relationships, known markers, perturbation responses, other external biological evidence? etc

- Applications

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